Python 3.15 Data Engineering Readiness Assessment

General Availability - Python 3.15.0 - tested 2026-10-10

92%

overall readiness | 13 PASS - 1 INCOMPAT - 1 BLOCKED - 2 SKIP - 0 FAIL (of 17 packages)

Executive summary

Python 3.15.0 was validated on macOS 27.0.1 ARM64 using uv. Of 17 packages probed, 13 passed and none failed outright. 4 did not pass: prefect (INCOMPAT); mlflow (SKIP); ray (BLOCKED); apache-airflow (SKIP).

Compared with 3.15.0rc2 (2026-09-05), overall readiness moved from 82% to 92%.

Test environment

Release	3.15.0 (General Availability)
Python runtime	3.15.0
Python build	cpython-3.15.0-macos-arm64-none
Platform	macOS 27.0.1 ARM64
Test date	2026-10-10
Tester	Dr. Ceasar Jackson Jr.
Suite version	1.2.0
Docker image	pyarrow-dataeng:py314
Environment consistent	NO - see limitations

Package results

Package	Status	Version	Notes
numpy	**PASS**	2.4.6	
pandas	**PASS**	3.0.3	
polars	**PASS**	1.41.2	
duckdb	**PASS**	1.5.3	
sqlalchemy	**PASS**	2.0.50	
pydantic	**PASS**	2.13.4	
matplotlib	**PASS**	3.10.9	
plotly	**PASS**	6.7.0	
jupyterlab	**PASS**	4.5.7	

Package	Status	Version	Notes
sqlite3	**PASS**	3.53.4	
pyarrow	**PASS**	26.0.0	
dask.dataframe	**PASS**	2026.7.1	
prefect	**INCOMPAT**	3.7.7	declared unsupported: requires Python >=3.10, <3.15, but `3.15.0` is installed
mlflow	**SKIP**	not installed	
ray	**BLOCKED**	unavailable	no cp315 wheels on PyPI
pyspark	**PASS**	4.1.2	via Docker py314 + OpenJDK 21
apache-airflow	**SKIP**	not installed	not installed

Changes since 3.15.0rc2

Change	Package	Was	Now	Version
Improvement	dask.dataframe	INCOMPAT	**PASS**	2026.7.1
Improvement	pyarrow	BLOCKED	**PASS**	26.0.0
Coverage change	pyspark	SKIP	**PASS**	4.1.2

14 package(s) unchanged.

Coverage changes (moving into or out of SKIP) are reported separately from compatibility changes: a package that was not measured before is not evidence of a regression or an improvement in Python itself.

Evidence and limitations

- Environment constraint violation: The package `prefect` requires Python >=3.10, <3.15, but `3.15.0` is installed
- A package that declares no support for this Python can still import and run; the suite classifies it INCOMPAT on declared support, not on smoke-test outcome.
- pyspark was exercised inside the Docker image tagged 'py314', not on the 3.15.0 runtime. It shows the package works in that container; it is not Python 3.15 evidence.
- The headline 92% counts those Docker-sourced results as PASS. Treating them as not measured (SKIP) instead, readiness would be 91%.
- mlflow was not measured (not installed); it is excluded from the readiness score.
- ray is BLOCKED: no cp315 wheels on PyPI.
- apache-airflow was not measured (not installed); it is excluded from the readiness score.

How readiness is calculated

Readiness = (PASS + 0.50 x BLOCKED + 0.25 x INCOMPAT) / (packages - SKIP), rounded to a whole percent. SKIP is excluded because an unmeasured package is neither compatible nor incompatible. The weights are this project's convention, not an industry standard, and the score summarises this suite's probes only; it is not a certification of any package or deployment.